

SAMSUN CARSAMBA, Turkey

WMO: 170310

Lat: **41.258N**Lon: **36.556E**Elev: **23**StdP: **14.68**Time Zone: **3.00 (E03)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.0	30.1	17.2	13.1	38.6	21.0	15.8	39.7	29.4	39.0	26.2	40.5	4.7	130	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	86.3	72.9	84.2	72.2	82.5	71.7	76.1	82.8	74.8	81.2	73.7	80.2	7.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	125.6	80.6	72.7	121.3	79.8	71.5	116.2	78.8	39.6	82.9	38.4	81.5	37.3	80.2	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.3	16.4	21.6	92.6	5.0	2.8	18.0	94.6	15.0	96.2	12.2	97.8	8.5	99.8		
(5)			20.5	77.6	4.7	1.5	17.1	78.7	14.4	79.6	11.8	80.5	8.4	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	44.3	45.4	48.0	52.9	61.2	69.0	73.5	74.6	68.3	61.1	53.2	46.8
	DBStd	12.13	7.27	7.40	6.80	6.11	5.14	4.01	3.06	3.14	4.18	5.38	6.13	6.72
	HDD50	708	208	166	123	32	0	0	0	0	0	1	35	143
	HDD65	3319	643	550	529	366	139	10	0	0	18	144	355	565
	CDD50	3724	30	36	60	119	349	571	728	763	548	344	132	44
	CDD65	860	0	0	1	4	22	131	263	298	116	22	2	1
	CDH74	6181	0	5	29	76	166	784	1921	2347	655	148	44	6
	CDH80	1128	0	0	5	20	30	132	336	476	93	32	4	0
Wind	WSAvg	6.2	7.0	7.0	6.6	5.8	5.2	5.7	6.5	6.6	5.6	5.3	5.9	6.5
Precipitation	PrecAvg	34.4	3.4	2.4	3.0	2.0	2.0	2.1	1.6	1.9	3.1	4.3	4.1	4.0
	PrecMax	46.7	4.9	4.3	5.7	3.0	7.3	4.4	4.8	7.0	6.7	9.0	10.0	6.6
	PrecMin	23.1	0.4	1.0	1.3	0.7	0.5	0.2	0.0	0.2	1.1	1.9	0.5	2.1
	PrecStd	4.9	1.2	1.0	1.1	0.7	1.6	1.1	1.3	1.8	1.4	1.8	2.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	73.4	78.3	82.5	83.2	88.0	88.0	89.2	86.4	84.3	79.7	73.1
		MCWB	50.9	54.0	56.4	61.4	66.8	70.7	74.0	74.7	70.0	65.5	58.8	54.9
	2%	DB	64.5	67.7	71.2	74.8	78.6	82.8	85.6	86.2	82.4	78.0	73.5	67.4
		MCWB	49.4	51.8	54.0	58.8	64.6	70.0	73.0	74.9	70.7	65.0	57.6	52.6
	5%	DB	60.9	62.9	65.8	69.4	74.8	80.7	83.6	84.2	79.9	73.5	69.5	62.8
		MCWB	48.1	49.7	52.2	56.7	63.8	69.7	72.5	73.7	69.8	64.5	56.4	50.6
	10%	DB	57.2	58.6	60.2	64.2	71.3	78.5	82.0	82.5	77.6	70.2	64.5	58.7
		MCWB	46.4	48.0	50.4	55.1	62.6	68.5	72.2	73.1	68.8	63.2	54.6	49.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	55.6	57.5	63.8	68.9	74.6	77.4	78.8	74.5	69.1	63.2	56.7
		MCDB	65.0	68.2	73.9	78.1	78.9	81.3	84.1	86.0	80.7	77.3	69.8	66.7
	2%	WB	50.4	52.8	54.9	59.8	66.8	72.8	75.8	76.6	72.4	67.1	60.1	54.1
		MCDB	61.1	65.0	67.7	71.9	75.5	79.5	82.3	83.6	79.2	74.0	67.8	63.3
	5%	WB	49.1	50.8	52.9	57.5	65.1	71.3	74.6	75.4	70.9	65.6	58.1	52.2
		MCDB	57.8	60.8	63.9	67.1	72.5	78.3	80.9	81.8	77.7	71.8	65.6	60.2
	10%	WB	47.5	48.8	50.8	55.5	63.4	69.8	73.4	74.3	69.5	64.1	56.4	50.6
		MCDB	54.4	56.6	58.9	63.2	69.7	76.4	79.7	80.7	76.1	69.8	63.5	56.8
Mean Daily Temperature Range	5% DB	MDBR	15.3	15.8	16.3	17.5	16.9	16.5	15.3	15.2	17.0	16.1	18.7	16.5
		MCDBR	22.1	25.7	28.6	29.3	23.9	21.0	18.7	17.9	20.9	23.3	27.8	24.6
		MCWBR	11.0	12.5	13.5	14.5	12.2	10.7	9.6	9.2	11.0	12.8	14.1	13.0
	5% WB	MCDBR	19.3	23.0	25.7	25.8	19.9	17.5	15.1	15.4	18.3	19.6	23.3	20.8
		MCWBR	10.6	12.1	13.0	13.6	10.9	9.8	9.0	9.0	10.4	11.5	13.4	12.1
Clear-Sky Solar Irradiance	taub		0.352	0.387	0.431	0.473	0.444	0.419	0.435	0.445	0.414	0.406	0.365	0.352
	taud		2.333	2.229	2.131	2.055	2.222	2.338	2.318	2.278	2.357	2.319	2.384	2.361
	Ebn at Noon		248	255	257	254	266	272	266	260	260	246	242	238
	Edh at Noon		28	36	44	51	44	40	40	41	35	33	27	25
All-Sky Solar Radiation	RadAvg		514	747	1073	1489	1818	2092	2066	1818	1408	933	645	457
	RadStd		58	91	84	154	180	166	106	105	97	84	75	56